Goal: Be able to solve (very basic) differential equations by integration.

Definition. For $y = f(x)$, a **differential equation** is an equation involving x, y, and the _____ of y.

Here are some examples (m and k are constants):

Equation	English Translation
$\frac{d^2y}{dx^2} = -9.81$	"Acceleration due to gravity is constant."
$\frac{dy}{dx} = ky$	"The rate a population grows is proportional to the amount of organisms in the population."
$\frac{d[y]}{dt} = -k[x_{(t)}][y_{(t)}]$	"The speed of a chemical reaction is proportional to the number of interactions between the molecules."
$m\frac{d^2y}{dx^2} + k\sin y = 0$	"The net force acting on an oscillating pendulum is 0."

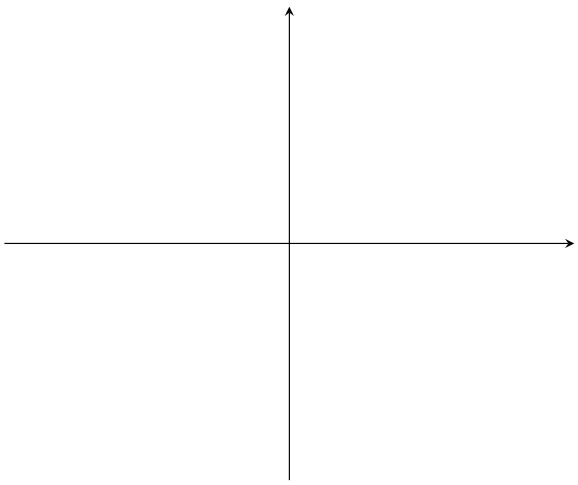
Example 1. Verify $y = 3e^{2x}$ is a solution to the differential equation $y' = 2y$.

Example 2. Solve the differential equation $y' = \frac{1}{x}$.

Definition. The **general solution** to a differential equation is a function that describes _____ of the differential equation.

Example 3. Find the general solution to $y' = 3x^2$

The general solution gives us a whole family of solutions to our differential equation.



Question: Is there a way to have a unique (i.e., only 1) solution?

Answer:

Definition. The extra condition above is an example of an _____. Usually, we specify when _____ (but that is not required).

IVPs

Definition. An **initial value problem (IVP)** is a differential equation combined with _____.

Example 4. The system

$$\begin{cases} y' = \sqrt{x} \\ y(0) = 1 \end{cases}$$

is an IVP. How can we solve it?

Definition. We say y is **(directly) proportional to** x if there is a constant k such that $y = kx$.

Example 5. The size of a population is governed by

$$\frac{dP}{dt} = \sqrt[3]{t}$$

where P is the population after t days (i.e., $\frac{dP}{dt}$ is proportional to $\sqrt[3]{t}$ with a constant coefficient of 1). If the initial population is 1000, find the population size after 8 days.

Multiple ICs

Fact: The have a unique solution, you need N ICs where N is the _____ in the differential equation.

Example 6. Solve the IVP:

$$\begin{cases} y''(t) = t - 3 \\ y'(1) = \frac{1}{2} \\ y(0) = 1 \end{cases}$$

Example 7. A ball is kicked off the ground with an initial velocity of $20\text{m}\,\text{s}^{-1}$. Describe the height of the ball as a function of time (assume gravity is the only force acting on the ball once it is kicked).

Example 8 (HW 27, Q6)**.** Solve the IVP:

$$\begin{cases} y''(x) = 4\sin x + \frac{10}{\pi^2}x \\ y'(\pi) = 4 \\ y\left(\frac{\pi}{2}\right) = -4 \end{cases}$$